

Home Lab Project Week 6, Day 4

Task:

Compare the Nmap results with the Wireshark observations. Document the reconnaissance footprint, save screenshots and PCAP evidence, and summarize how network defenders could recognize the activity.

Comparing Nmap Results with Wireshark Observations

Engagement: Home Lab Internal Network Assessment **Lab log:** Week 6, Day 4 **Target:** Ubuntu 192.168.78.5 · **Tester:** Kali Linux 192.168.78.3
Prepared by: [Your name], DAE Cybersecurity Program **Date:** [Fill in]

1. Objective

My objective was to compare the Nmap scan results with what I observed in the Wireshark capture, document the reconnaissance footprint the scan produced, save the supporting evidence (screenshots and PCAP), and summarize how a network defender could recognize this activity.

2. Comparing the Two Views

Nmap reports its *conclusions* about the target; Wireshark shows the *raw packets* behind those conclusions. When I line them up, every Nmap result is explained by a specific packet exchange in the capture.

Nmap result

Corresponding Wireshark observation

Host is up	The target responded to probes (low latency, replies received)
Port 21/tcp open (ftp)	Target sent a SYN-ACK from port 21
Port 23/tcp open (telnet)	Target sent a SYN-ACK from port 23
998 closed ports	Target sent RST-ACK replies from those ports
SYN (half-open) scan	For open ports, Kali sent a RST instead of completing the handshake
Scan of ~1,000 ports	A burst of ~1,000 SYN packets from Kali in a few seconds

Conclusion from the comparison: the two tools fully corroborate each other. Nmap's summary ("21 and 23 open, 998 closed") is exactly what the packet-level evidence shows — the open ports answered with SYN-ACK, the closed ports answered with RST-ACK, and the scan never completed a handshake. This confirms the Nmap results are accurate and shows *why* they are true at the protocol level.

3. Reconnaissance Footprint

The "footprint" is the observable trace the scan left behind — everything a defender could use to detect and attribute the activity.

On the network:

- A single source IP, **192.168.78.3** (Kali), identifiable as the origin of the scan.
- Roughly **1,000 outbound SYN packets** to one host, one per port.
- All within a very **short time window** (seconds).

- A matching **burst of ~1,000 RST-ACK packets** generated by the target.
- Two **SYN-ACK** responses revealing the open services (21 and 23).
- Many **half-open, never-completed** connections — a high ratio of SYN packets to actual data sessions.

On the target host:

- The scan touched ~1,000 ports, which may appear in firewall or connection logs as a wave of connection attempts from one IP.
- The MAC address **08:00:27:E3:80:BC** ties the traffic to the specific VM, and the source IP identifies the scanning machine.

What the footprint reveals: the volume, timing, and one-to-many pattern make this unmistakably reconnaissance rather than normal use. The attacker's IP is exposed, and the intent — mapping exposed services — is clear from the behavior.

4. Evidence

PCAP: `week6-day2-nmap-scan.pcapng` (saved via **File** → **Save As** in Wireshark).

Screenshots to save:

- Nmap results showing ports 21 and 23 open, 998 closed.
- Wireshark SYN flood from Kali (`tcp.flags.syn==1 && tcp.flags.ack==0`).
- SYN-ACK replies from ports 21 and 23 (`tcp.flags.syn==1 && tcp.flags.ack==1`).
- RST-ACK replies from closed ports (`tcp.flags.reset==1`).
- **Statistics** → **Conversations** showing one source fanning out to ~1,000 ports.

Placing the Nmap screenshot next to the Wireshark screenshots side by side is the clearest way to show the comparison in the report.

5. How Network Defenders Could Recognize This Activity

A defender would identify this scan by its pattern rather than any single packet:

- **One source, many destinations, short time.** ~1,000 ports on one host from a single IP in seconds is not normal traffic — it is the defining shape of a port scan.
- **Many incomplete connections.** A high ratio of SYN packets to completed, data-carrying sessions signals scanning rather than legitimate use.
- **A burst of RSTs from the target.** The host generating ~1,000 resets at once is itself anomalous.
- **IDS/IPS detection.** Tools like Snort or Suricata count connection attempts from one source across many ports within a time window and alert when a threshold is crossed.
- **Firewall and SIEM correlation.** Logs showing one IP hitting many ports stand out immediately and can be correlated automatically.

Typical defender response: alert on and log the source IP, block or rate-limit it (e.g. with a firewall rule or fail2ban), close unnecessary ports to shrink the attack surface, and segment the network so a scanner cannot freely probe other systems.

6. Observations & Conclusion

- The Nmap results and Wireshark capture agree completely — the packet evidence explains and confirms every conclusion Nmap reported.

- The scan left a clear reconnaissance footprint: an identifiable source, a large one-to-many SYN burst, and many half-open connections.
 - This footprint is exactly the kind of pattern network defenders and intrusion-detection systems are designed to catch, which is why reconnaissance is often the first stage where an attacker can be detected.
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7. Evidence Log

- [] [week6-day2-nmap-scan.pcapng](#) — scan capture
 - [] Nmap results screenshot (ports 21, 23 open; 998 closed)
 - [] Wireshark SYN flood screenshot
 - [] Wireshark SYN-ACK (open ports) screenshot
 - [] Wireshark RST-ACK (closed ports) screenshot
 - [] **Statistics** → **Conversations** screenshot (one-to-many fan-out)
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This document records authorized security testing I performed in a controlled lab environment.